

Charles T. Clark

Engineering Portfolio: www.charlesclark.me

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OBJECTIVE

Third year mechanical engineering student seeking Co-op for Spring 2027 and Internship for Summer 2027

EDUCATION

Virginia Tech

Bachelor of Science, Mechanical Engineering Candidate

Blacksburg, VA

Expected Graduation: May 2028

- GPA: 3.84 | Dean's List (All Semesters)
- Mechanical Engineering Undergraduate Student Ambassador
- Raymond & Shirley Lynn Merit Scholarship Recipient
- Relevant Courses: Dynamics, Thermodynamics, Differential Eq, Mechanics of Bodies, Electrical Theory

EXPERIENCE & PROJECTS

Culligan International

Mechanical Engineering Design Intern

May 2026 – Aug 2026

Rosemont, IL

- Extensive usage of SolidWorks and FEA to design, develop and build water treatment components
- Designed new float system with injection molded parts for a residential water softener system
- Developed mounting brackets to improve water pressure for commercial hydraulic system
- Created and presented a portable field training "suitcase" prototype that secured \$75,000 in funding
- Drafted technical engineering drawings for 5 commercial water softeners and assemblies

Pulse Jet Design Team

Founder and President

May 2025 – Present

Blacksburg, VA

- Founded a design team with 8 members that prototypes and builds pulse jet engines
- Designed in SolidWorks and fabricated a steel, propane-fueled pulse jet that produced approximately 300 Newtons of peak static thrust during bench testing
- Created a testing bench to safely and accurately get thrust, temperature, and fuel efficiency readings

Virginia Tech Engineering Education Department

Accessibility Design Project Leader

August 2024 – May 2025

Blacksburg, VA

- Led a team to design and prototype a telescoping crutch, reducing storage space by 66%
- Utilized CAD and rapid prototyping to create models and show the device's capabilities

Automotive Restoration

Independent Project

August 2020 – Present

Libertyville, IL

- Rebuilt and restored two vehicles, a 2001 Jeep Cherokee and a 1987 Jeep Comanche.
- Used welding, machining, and diagnostic problem solving to restore vehicles to working conditions.

HONORS & CERTIFICATIONS

Eagle Scout | *Boy Scouts of America*

July 2023

Certified SolidWorks Professional (CSWP) | *Dassault Systèmes*

July 2026

Finite Element Analysis (FEA) Certification | *GoEngineer*

July 2026

TECHNICAL SKILLS

CAD and Analysis: SolidWorks (CSWP), Finite Element Analysis (FEA), Engineering Drawings, MATLAB

Product Development: Rapid Prototyping, Injection Molded Part Design, Design Validation

Manufacturing & Fabrication: MIG/Stick Welding, 3D Printing, CNC Machining, Forklift Certification